

Chapter 9: Vectors in Two and Three Dimensions

Name: _____

Date: _____

Instructions: Answer all questions. Show complete work and justify steps. Give exact values when possible; otherwise round angles to the nearest degree. Use $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and component notation as appropriate. Each problem begins on a new page.

1. Let $\mathbf{u}$ be the vector with initial point $P(-3, 2)$ and terminal point $Q(5, -4)$. (8)

(a) Express $\mathbf{u}$ in terms of $\mathbf{i}$ and $\mathbf{j}$.

(b) Find $\|\mathbf{u}\|$ and the direction angle of $\mathbf{u}$.

2. Let $\mathbf{u} = \langle 1, 3 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$. (10)
- (a) Compute $\mathbf{u} - 2\mathbf{v}$ and $\|\mathbf{u} + \mathbf{v}\|$.

- (b) Compute $\mathbf{u} \cdot \mathbf{v}$ and determine whether $\mathbf{u}$ and $\mathbf{v}$ are perpendicular.

3. Let $\mathbf{u} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = -\mathbf{i} + 2\mathbf{j} + \mathbf{k}$. (12)

(a) Find $\mathbf{w}$ if $\mathbf{u} - 2\mathbf{w} = \mathbf{v}$.

(b) Find the angle between $\mathbf{u}$ and $\mathbf{v}$ (nearest degree).

4. Let $A(1, -4, 2)$ and $B(4, 2, -1)$. (10)
- (a) Find the distance AB and the midpoint of $\overline{AB}$.

- (b) Find the equation of the sphere for which $\overline{AB}$ is a diameter.

5. Let $\mathbf{a} = \langle 2, -1, 4 \rangle$ and $\mathbf{b} = \langle -3, 5, 1 \rangle$. (10)

(a) Compute $\mathbf{a} \times \mathbf{b}$.

(b) Find the area of the triangle with vertices at $O(0, 0, 0)$, A with position vector $\mathbf{a}$, and B with position vector $\mathbf{b}$.

6. (a) Find parametric equations for the line through $(2, -1, 3)$ parallel to $\langle 4, 0, -2 \rangle$. (12)

(b) Find an equation of the plane through the points $P(1, 0, 2)$, $Q(3, -1, 1)$, and $R(2, 2, 0)$.

(c) Find the angle between the plane from part (b) and the plane $x + 2y - z = 1$ (nearest degree).

7. Find the work done by the constant force $\mathbf{F} = \langle 3, -5, 4 \rangle$ in moving an object from $(0, 0, 0)$ to $(2, -1, 3)$ along a straight line. (8)

8. Let $\mathbf{u} = \langle 6, -2, 3 \rangle$ and $\mathbf{v} = \langle 3, 0, 4 \rangle$. (10)

(a) Find the scalar component of $\mathbf{u}$ along $\mathbf{v}$.

(b) Find $\text{proj}_{\mathbf{v}} \mathbf{u}$.

9. A current in a straight canal flows due south at 2 m/s. A swimmer can move at 1.5 m/s relative to the water. (12)

(a) If the swimmer aims due east, find the true velocity vector (components in the east/north coordinate system), the true speed, and the bearing.

(b) What heading (nearest degree) should the swimmer choose to end up moving due east relative to the ground? What will the resulting speed be?

10. Let $\mathbf{u} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, and $\mathbf{w} = \mathbf{j} + \mathbf{k}$. (8)

(a) Compute $2\mathbf{u} + 3\mathbf{v}$ and $\|\mathbf{u}\|$.

(b) Compute $\mathbf{u} \times \mathbf{v}$ and the scalar triple product $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{w})$.